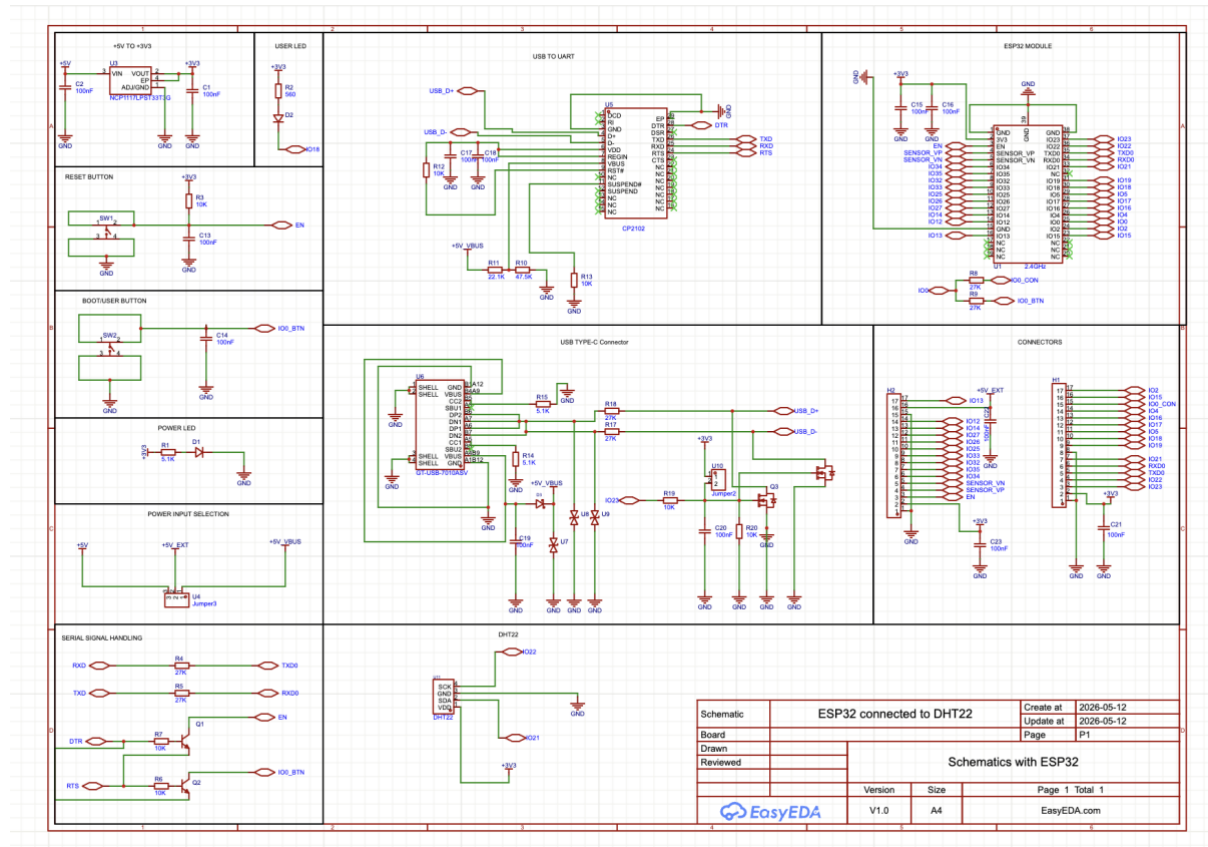


Embedded Systems & IoT April -July 2026

Lab Exercise: Schematics with ESP32

ESP32 connected to DHT22



No.	Footprint	Comment	Designator	Primary Category	Secondary Category	Supplier
1	C0603	100nF	C1,C2,C13,C14,C15,C16,C17,C18,C19,C20,C21,C22,C23	Capacitors	Multilayer Ceramic Capacitors MLCC - SMD/SMT	
2	HDR-TH_2P-P2.54-V-M-1	Jumper2	U10			LCSC
3	HDR-TH_3P-P2.54-V-M-A	Jumper3	U4			LCSC
4	HDR-TH_17P-P2.54-V-M	X6511 WV-17H-C60D30	H1,H2	Connectors	Pin Headers	LCSC
5	QFN-28_L5.0-W5.0-P0.50-TL-EP3.2	CP2102	U5			LCSC
6	R0603	5.1K,10K,2.1K,27K,4.7K,560	R1,R2,R3,R4,R5,R6,R7,R8,R9,R10,R11,R12,R13,R14,R15,R17,R18,R19,R20	Resistors	Chip Resistor - Surface Mount	
7	SENSOR-TH_HAIGU_DHT22	DHT22	U11	Sensors	Temperature and Humidity Sensor	LCSC
8	SMA_L4.4-W2.8-LS5.4-RD	1N4007	D1,D2			

9	SOD-123_L2.8-W1.8-LS3.7-RD	B5819W-TP	D3	Diodes	Schottky Diodes	LCSC
10	SOD-523_L1.2-W0.8-LS1.6-BI	LES5D5.0CT1G-ES	U7,U8,U9	Circuit Protection	ESD and Surge Protection (TVS/ESD)	LCSC
11	SOT-23-3_L2.9-W1.3-P1.90-LS2.4-BR	LM-2N7002LT1G,PBSS4240	Q1,Q2,Q3,Q4	Transistors/Thyristors	Bipolar (BJT),MOSFETs	LCSC
12	SOT-223_L6.5-W3.5-P2.30-LS7.0-BR	NCP1117LPST33T3G	U3	Power Management (PMIC)	Voltage Regulators - Linear, Low Drop Out (LDO) Regulators	LCSC
13	SW-SMD_4P-L5.2-W5.2-P3.7-LS6.4-1	HX 5.2x5.2x1.5H SWITCH TACTILE	SW1,SW2	Switches	Tactile Switches	LCSC
14	USB-C-SMD_G-SWITCH_GT-USB-7010ASV	GT-USB-7010ASV	U6	Connectors	USB Connectors	LCSC
15	WIFI-SMD_ESP32-WROOM-32E	2.4GHz	U1	IoT/Communication Modules	WiFi Modules	LCSC